System Design :→

↳ whatsapp

1) websockets
2) Presence (online or not)
3) Happy case msg delivery
4) offline msg delivery
5) ordering ensure
6) Read or not
7) Group chats, multi-device sync
8) typing indicators

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functional Req
↳ all the API's
↳ out of scope

non functional
↳ latency < 500 ms
↳ Gaurantee msg delivery
↳ no of users → qps → query per second
↳ storage
↳ fault tolerant

core entities → what we keep in the table
user, msgs, chat, client → device
multiple clients per user

Connections

How client → Yes/No on Each

Latency → Yes → Freq Bi-dir → Peer to Peer Audio/Video → Yes → webRTC
↓ websocket
↓ SSL
No → polling

In our case → we need websocket
what is websocket connection
whatsApp usually use TLS
why not HTTP → server to client (N/A) in HTTP
we need persistent Bi-directional

Since Bi-directional → two types of call

1) C → Server
→ send msg
→ create grp chat
→ modify participants
→ create attachment

2) Server → client
receive msg
client update

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if B is online, do we not store msgs in dB
↳ makes sense because we want to support low latency → but if I change to a new phone how do I retrive my msgs

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with WS, the Load balancer cannot route to any server-chat, unlike HTTP, where things were stateless, there we use L-7 LB, here we need a L-4 LB
with HTTP → web-servers are stateless

[Least connection algo]

(Redis pub/sub) → fast → server → WS → discovery